

Simultaneous diagonalization of nearly commuting Hermitian matrices: do-one-then-do-the-other

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Commuting Hermitian matrices may be simultaneously diagonalized by a common unitary matrix. However, the numerical aspects are delicate. We revisit a previously rejected numerical approach in a new algorithm called ‘do-one-then-do-the-other’. One of two input matrices is diagonalized by a unitary similarity, and then the computed eigenvectors are applied to the other input matrix. Additional passes are applied as necessary to resolve invariant subspaces associated with repeated eigenvalues and eigenvalue clusters. The algorithm is derived by first developing a spectral divide-and-conquer method and then allowing the method to break the spectrum into, not just two invariant subspaces, but as many as safely possible. Most computational work is delegated to a black-box eigenvalue solver, which can be tailored to specific computer architectures. The overall running time is a small multiple of a single eigenvalue-eigenvector computation, even on difficult problems with tightly clustered eigenvalues. The article concludes with applications to a structured eigenvalue problem and a highly sensitive eigenvector computation.

Keywords: simultaneous diagonalization; joint diagonalization; nearly commuting; eigenvalue decomposition; spectral divide-and-conquer.

1. Introduction

Commuting Hermitian matrices A and B may be simultaneously diagonalized by a unitary similarity. That is, there exists a unitary matrix U such that $U^H A U =: \Lambda$ and $U^H B U =: M$ are both diagonal (Horn & Johnson, 1990, p. 235). However, numerical computation is complicated by potential ill conditioning of eigenvectors. Bunse-Gerstner *et al.* (1993) illustrate the difficulty by showing how a natural *diagonalize-one-then-diagonalize-the-other* method can fail. The method diagonalizes A then applies its eigenvectors to B . If A has any multiple eigenvalues, then some of its eigenvectors are underdetermined and may not be sufficient for diagonalizing B . Hence, the method shifts its attention from A to B in an attempt to resolve every eigenvector. Bunse-Gerstner *et al.* show that this method fails in the presence of clusters of very close eigenvalues, and then they shift focus to a slower Jacobi-like algorithm.

In this article, we revisit the ‘one-then-the-other’ idea and develop a related, but new algorithm for simultaneous diagonalization that we call *DODO*, short for *do-one-then-do-the-other*. The method is backward stable, and it is fast and flexible: the computationally-intensive stages of the algorithm are delegated to a black-box eigenvalue solver, which can be tailored to specific computer architectures. Numerical experiments show the execution time at a small multiple of a single eigenvalue-eigenvector computation.

Ultimately, we develop two algorithms in this article. The first is a spectral divide-and-conquer method that simultaneously reduces A and B to block-diagonal form and then recursively reduces the blocks until both matrices are diagonal. The second is the new DODO algorithm, which diagonalizes

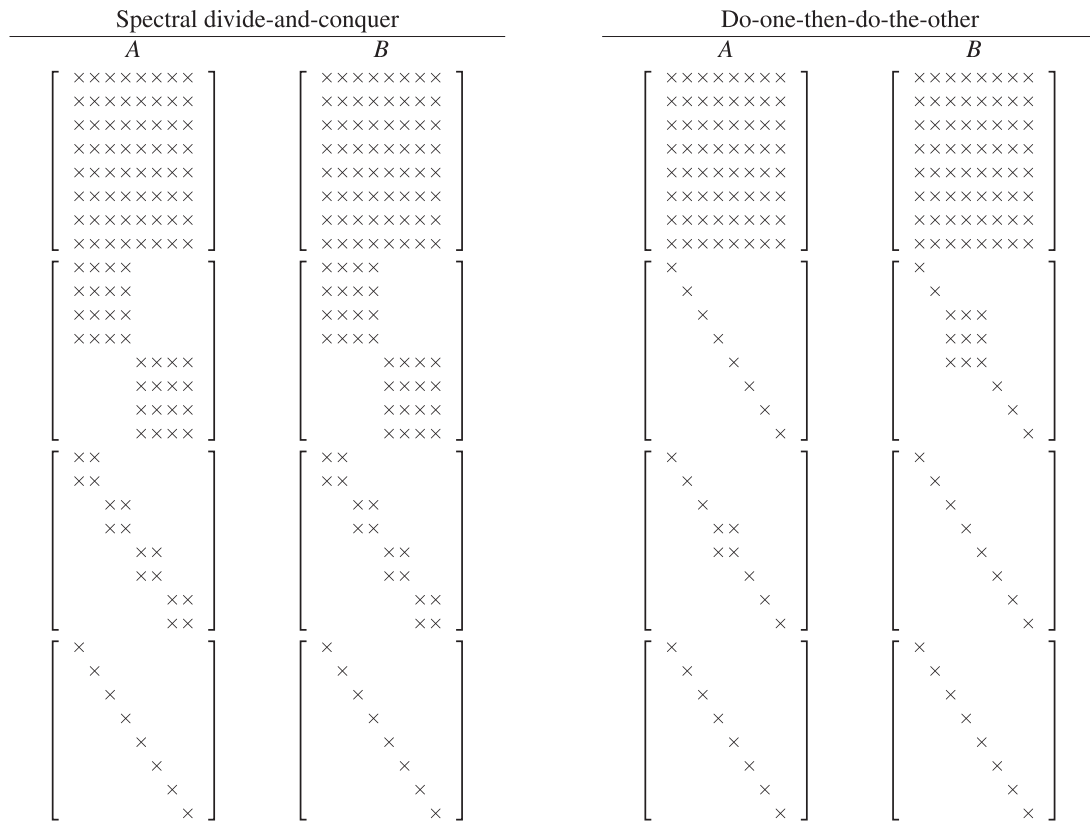


FIG. 1. Two algorithms for simultaneous diagonalization. In an ideal run of spectral divide-and-conquer, the number of blocks on the diagonal is doubled with each recursive step. With do-one-then-do-the-other, the diagonal blocks are determined adaptively based on eigenvalue clusters; when all eigenvalues are well separated, a single step is sufficient.

one matrix before moving to the other, refining things only when a small eigenvalue gap is detected. The spectral divide-and-conquer method is convenient for theoretical analysis because it concentrates on a single eigenvalue gap at a time. The DODO method is advantageous in practice because it more aggressively zeroes off-diagonal entries. Runs of the two algorithms are illustrated in Fig. 1.

A proof of backward stability is presented for the spectral divide-and-conquer algorithm, and an outline of the very similar proof for the DODO algorithm is provided as well. Numerical experiments verify backward stability and show that the speed of DODO depends modestly on the number and size of eigenvalue clusters, running nearly as fast as a single eigenvalue computation when eigenvalues are well separated. Thus, we have an algorithm that is backward stable like the prior Jacobi-like algorithm, but that is guaranteed to terminate and that automatically benefits from newer, faster methods for the symmetric eigenvalue problem.

1.1 Problem statement

Hermitian matrices A and B are δ -commuting if there exist Hermitian matrices ΔA and ΔB of 2-norm $\leq \delta$ for which $A + \Delta A$ and $B + \Delta B$ commute. Our main problem is to simultaneously diagonalize such matrices with a backward error proportional to δ :

PROBLEM 1 Suppose A and B are δ -commuting n -by- n Hermitian matrices. Compute a unitary matrix $\hat{U}$ that simultaneously diagonalizes both inputs, specifically

$$\hat{U}^H(A + \Delta A)\hat{U} = \Lambda, \quad \hat{U}^H(B + \Delta B)\hat{U} = M,$$

for some backward errors satisfying $\|\Delta A\| \leq c_n \delta$ and $\|\Delta B\| \leq c_n \delta$. The coefficient c_n may grow slowly with n , but may not depend on δ .

Throughout the article, $\|\cdot\|$ denotes the 2-norm, i.e., the largest singular value.

Note that the ΔA and ΔB in the problem statement are different from the ΔA and ΔB in the definition of δ -commuting; we do not expect to compute the nearest commuting matrix pair exactly, but we do seek to be within a constant factor c_n for any given matrix size n .

1.1.1 Scaling. Problem 1 is stated in the language of absolute error. In practice, one may wish to rescale the matrices A and B before solving. We discuss two examples.

First, consider a model in which A and B are observations, subject to measurement error, of exactly commuting matrices $\mathcal{A}$ and $\mathcal{B}$:

$$\mathcal{A}\mathcal{B} = \mathcal{B}\mathcal{A}, \quad \mathcal{A} = A + \delta_1 E_1, \quad \mathcal{B} = B + \delta_2 E_2, \quad \|E_1\| = \|E_2\| = 1.$$

Then A and B are $\max\{\delta_1, \delta_2\}$ -commuting, and, without rescaling, a solution to Problem 1 produces backward errors bounded by $c_n \max\{\delta_1, \delta_2\}$. This seems weaker than necessary if δ_1 and δ_2 are incommensurate. To correct this, set $\delta = \sqrt{\delta_1 \delta_2}$ and define $\tilde{A} = (\delta/\delta_1)A$ and $\tilde{B} = (\delta/\delta_2)B$. Because $\tilde{A}$ and $\tilde{B}$ are δ -commuting, a solution to Problem 1 produces

$$\hat{U}^H(\tilde{A} + \Delta \tilde{A})\hat{U} = \tilde{\Lambda}, \quad \hat{U}^H(\tilde{B} + \Delta \tilde{B})\hat{U} = \tilde{M}$$

with $\|\Delta \tilde{A}\|, \|\Delta \tilde{B}\|$ bounded by $c_n \delta$. Scaling the first equation by δ_1/δ and the second by δ_2/δ gives

$$\hat{U}^H(A + \Delta A)\hat{U} = \Lambda, \quad \hat{U}^H(B + \Delta B)\hat{U} = M$$

with $\|\Delta A\| = (\delta_1/\delta)\|\Delta \tilde{A}\| \leq c_n \delta_1$ and $\|\Delta B\| = (\delta_2/\delta)\|\Delta \tilde{B}\| \leq c_n \delta_2$. Note that the backward errors $\Delta A, \Delta B$ are proportional in norm to the original measurement errors $\delta_1 E_1, \delta_2 E_2$. Of course, to execute this strategy, estimates are needed for δ_1 and δ_2 .

Our second rescaling focuses on relative error. If A and B are δ -commuting, then the normalized matrices $\tilde{A} := A/\|A\|$ and $\tilde{B} := B/\|B\|$ are $\max\{\delta/\|A\|, \delta/\|B\|\}$ -commuting, and a solution to Problem 1 commits backward errors with norms $\|\Delta \tilde{A}\|$ and $\|\Delta \tilde{B}\|$ bounded above by $c_n \max\{\delta/\|A\|, \delta/\|B\|\}$. Reversing the scaling, this achieves

$$\hat{U}^H(A + \Delta A)\hat{U} = \Lambda, \quad \hat{U}^H(B + \Delta B)\hat{U} = M$$

with

$$\frac{\|\Delta A\|}{\|A\|}, \frac{\|\Delta B\|}{\|B\|} \leq c_n \max \left\{ \frac{\delta}{\|A\|}, \frac{\delta}{\|B\|} \right\}.$$

Thus, we have derived a bound on relative error from a solution to Problem 1.

Because the user may wish to scale inputs explicitly, e.g., to balance measurement errors between A and B , our algorithms perform no automatic rescaling, and the analysis focuses on the fundamental quantities $\|\Delta A\|$ and $\|\Delta B\|$, from which other bounds can be derived.

1.2 Background

1.2.1 Prior work and applications. The major existing method for simultaneous diagonalization is the Jacobi-like method by [Bunse-Gerstner et al. \(1993\)](#). See also the references therein and [Cardoso & Souloumiac \(1996\)](#). The Jacobi-like method has received widespread use as part of the JADE algorithm for Independent Component Analysis ([Cardoso & Souloumiac, 1993](#)). It is backward stable and locally quadratically convergent, but its arithmetic operation count is high, its convergence is not guaranteed for some known examples (see section 4.4), and its termination criterion depends on *a priori* knowledge of the input matrices' deviation from commutativity.

Our spectral divide-and-conquer method is motivated by existing algorithms for the symmetric eigenvalue problem; see the work of [Nakatsukasa and Higham \(2013\)](#) and their references.

Simultaneous diagonalization has application to several structured eigenvalue problems ([Bunse-Gerstner et al., 1992](#)). The simultaneous diagonalization of several commuting Hermitian matrices, also known as *joint diagonalization*, has additional applications, e.g., beamforming in wireless communications ([Cardoso & Souloumiac, 1993](#)) and multimodal shape analysis (see, e.g., [Eynard et al., 2015](#)). Although the present article is limited to a pair of commuting matrices, we expect the analysis to be useful in the future consideration of three or more matrices.

1.2.2 Nearly commuting versus almost commuting. The problem of simultaneous diagonalization is well posed for Hermitian matrices A and B that *nearly commute*. This means that there exist Hermitian matrices ΔA and ΔB of small norm for which $A + \Delta A$ and $B + \Delta B$ commute. A related, but different notion is *almost commutativity*, which means that $AB - BA$ has small norm, a less restrictive criterion. This article is largely about near commutativity, but the relationship with almost commutativity is important the proof of Theorem 2.9. Appendix A discusses the relationship between near and almost commutativity.

1.2.3 The $\sin \theta$ theorem of Davis and Kahan. In the proofs of backward stability, we must show that certain invariant subspaces are computed accurately enough to block-diagonalize a pair of matrices. The $\sin \theta$ theorem of [Davis and Kahan \(1969; 1970\)](#) is our basic tool. Appendix B provides a corollary of the $\sin \theta$ theorem and discusses its significance.

1.2.4 Computational kernels. We require computational kernels for orthogonal transformation, invariant subspaces and the symmetric eigenvalue problem:

$$\begin{aligned} Q^T X &= Y + E, \\ X &= V \begin{bmatrix} Y_{11} & \\ & Y_{22} \end{bmatrix} V^H + E, \\ X &= U \Lambda U^H + E. \end{aligned}$$

In each instance, E is an error term satisfying the bound

$$\|E\| \leq c_n \|X\| u, \quad (1.1)$$

in which c_n is a slowly growing function of the matrix size n and u is unit roundoff.

We prefer to avoid juggling both δ and u in the notation. For δ -commuting matrices A and B stored in finite precision, we expect that typically

$$\delta \geq \max\{\|A\|, \|B\|\} u.$$

To be safe, let δ' be the larger of δ or $\max\{\|A\|, \|B\|\} u$. Then $u \leq \delta' / \max\{\|A\|, \|B\|\}$ for sure, and the three error bounds above imply $\|E\| \leq c_n \delta'$ when $X = A$ or $X = B$. For the remainder of the article, we assume

$$\|E\| \leq c_n \delta \quad (1.2)$$

for each of the three operations, with the understanding that δ has been replaced by δ' if necessary.

1.3 Organization

The article is organized into three major parts: Section 2 presents and analyzes the spectral divide-and-conquer method; section 3 presents and analyzes the DODO method; and section 4 shares results from numerical experiments, including applications to a structured eigenvalue problem and a highly ill-conditioned eigenvalue problem.

2. Spectral divide-and-conquer method

In this section, we develop our spectral divide-and-conquer method for simultaneously diagonalizing Hermitian matrices A and B that nearly commute.

The method starts by finding a linear subspace that is nearly invariant for both A and B . Orthonormal bases for the invariant subspace (V_1) and its orthogonal complement (V_2) are chosen, and the unitary matrix $V = \begin{bmatrix} V_1 & V_2 \end{bmatrix}$ is applied to both inputs:

$$V^H A V = \begin{bmatrix} V_1^H A V_1 & 0 \\ 0 & V_2^H A V_2 \end{bmatrix}, \quad V^H B V = \begin{bmatrix} V_1^H B V_1 & * \\ * & V_2^H B V_2 \end{bmatrix}. \quad (2.1)$$

The off-diagonal blocks of the second matrix are small, but likely nonzero when the input matrices do not perfectly commute. The algorithm seeks a large eigenvalue gap to control the norms of these off-diagonal blocks, and we prove a bound that allows the algorithm to safely zero the entries.

The method then simultaneously diagonalizes the blocks on the main diagonal, recursively.

The spectral divide-and-conquer method is presented as Algorithm 1. The algorithm accepts Hermitian matrices A and B and computes a unitary matrix U that nearly diagonalizes both inputs. Parameters α and β are tolerances that control trade-offs between speed and stability; the details are explained below. The notation $s(A)$ refers to the eigenvalue *spread* of A : $s(A) = \lambda_{\max}(A) - \lambda_{\min}(A)$.

Because larger eigenvalue gaps are associated with better-conditioned invariant subspaces, the first priority is to find a moderately large eigenvalue gap. In step 1, the algorithm selects one of the input matrices A or B on the basis that it is guaranteed to have a relatively large gap in its spectrum. Generally,

ALGORITHM 1 Spectral divide-and-conquer

Input: Hermitian A and B ; $\alpha, \beta \in (0, 1]$.

Output: Unitary U and diagonal Λ and M such that $U^H A U \approx \Lambda$ and $U^H B U \approx M$.

1. Select a matrix with a relatively large spectral spread.
 - (a) Case 1: $s(A) = s(B) = 0$. A and B are multiples of the identity matrix. Terminate with $U = I$, $\Lambda = A$ and $M = B$.
 - (b) Case 2: $s(A) \geq \beta s(B)$. Set $C = A$.
 - (c) Case 3: $s(B) \geq \beta s(A)$. Set $C = B$.
 2. Find an eigenvalue gap of width $(\alpha/n)s(C)$ in the spectrum of C .
 3. Compute invariant subspaces V_1 and V_2 of C corresponding to the eigenvalues on either side of the gap.
 4. Simultaneously block-diagonalize. Compute $V_i^H A V_i$ and $V_i^H B V_i$, $i = 1, 2$.
 5. Solve subproblems. Recursively compute $U_i^H V_i^H A V_i U_i \approx \Lambda_i$ and $U_i^H V_i^H B V_i U_i \approx M_i$ for $i = 1, 2$.
 6. Combine. Set $U = [V_1 U_1 \quad V_2 U_2]$, $\Lambda = \text{diag}(\Lambda_1, \Lambda_2)$ and $M = \text{diag}(M_1, M_2)$.
-

it is safer to choose the matrix with a larger spectral spread, but to provide some flexibility, tolerance β is introduced; if the spreads are within a factor of β of each other, then either spectrum may be chosen. (See Remark 2.1.)

In step 2, an eigenvalue gap is chosen from the selected spectrum. Within the selected spectrum $\sigma(C)$, there must be an eigenvalue gap of width at least $(1/n)s(C)$ because the average of the gap sizes is $s(C)/(n-1)$. A gap of width at least $(\alpha/n)s(C)$ is chosen. It is always possible to set $\alpha = 1$, but a smaller tolerance may expand the set of possibilities. (See again Remark 2.1.)

After the eigenvalue gap is located, corresponding invariant subspaces are computed in step 3.

In step 4, the input matrices are block diagonalized as in (2.1), and the off-diagonal blocks are implicitly zeroed.

In step 5, the recursive subproblems are solved, and in step 6, the subproblems are recombined.

The backward stability of the algorithm rests on a justification for replacing off-diagonal blocks by zeros. If the spectrum of A is chosen in step 1 and $V = [V_1 \quad V_2]$ identifies invariant subspaces for A , then $V^H A V$ is block diagonal by design. The question is whether the off-diagonal blocks of $V^H B V$ are small enough to discard. Below, we prove that this is so. The idea is roughly as follows. If A has a reasonably large spread relative to δ (the value for which A and B are δ -commuting), then there must be an eigenvalue gap also reasonably large relative to δ (specifically of width at least $(\alpha/n)s(A)$). Then V_1 and V_2 are accurate approximations to invariant subspaces that are common to A and B and hence simultaneously block-diagonalize both matrices. The other possibility is that only tiny eigenvalue gaps are present, and therefore V_1 and V_2 likely contain garbage. However, this can only happen if $s(A)$ and $s(B)$ are tiny, in which case both matrices are nearly equal to multiples of the identity matrix. Such matrices have most of their mass along the main diagonal, regardless of the choice of basis represented by V . (See the proof of Lemma 2.2 below.)

The formal analysis does not immediately split into two cases as the previous paragraph may suggest, but rather it bounds $\|V_2^H B V_1\|$ by a product of two factors, one reflecting the eigenvalue gap and the other reflecting concentration of mass along the diagonal. As the spread decreases, one factor grows while the other shrinks, so that their product remains small and the error remains uniformly bounded. (See the comment following Lemma 2.2.) The error analysis culminates in Theorem 2.10 below, which establishes backward stability.

REMARK 2.1 For the purpose of theoretical analysis, we leave two partially specified steps in Algorithm 1: (1) if $\beta s(B) \leq s(A) \leq s(B)$ or $\beta s(A) \leq s(B) \leq s(A)$, then either step 1(b) or step 1(c) may be executed; (2) if there are multiple eigenvalue gaps of sufficient width in step 2, then any of them may be selected. We endeavor to prove the algorithm's effectiveness regardless of how these choices are made. In practice, this flexibility can be leveraged. First, determining which of two matrices has a larger spread can be nontrivial; establishing that $s(A) \geq \beta s(B)$ for some positive β can be much cheaper than establishing $s(A) \geq s(B)$. Secondly, one may prefer to split at an eigenvalue gap near the median of the spectrum even if it is not the largest eigenvalue gap; this produces a more balanced partition, accelerating the divide-and-conquer scheme. (At the same time, one should be mindful of the roles of α and β in the upcoming error bounds.) These partially specified steps are considered again when we introduce our second numerical method; see particularly section 3.4.

2.1 Examples

A couple of examples illustrate the spectral divide-and-conquer algorithm.

The 1993 article by Bunse-Gerstner, Byers and Mehrmann considers the example

$$A = \begin{bmatrix} 1-\varepsilon & 0 & 0 & 0 \\ 0 & 1+\varepsilon & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1-\varepsilon & 0 \\ 0 & 0 & 0 & 1+\varepsilon \end{bmatrix}.$$

This example is tricky because the invariant subspace of A spanned by $(1, 0, 0, 0)$ is not at all an invariant subspace for B . Likewise, the invariant subspace of B spanned by $(0, 0, 1, 0)$ is not an invariant subspace for A . Still, the unitary matrix $\text{diag} \left(\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \right)$ nearly diagonalizes both A and B .

Let's run the spectral divide-and-conquer method on A, B . The matrices have identical eigenvalues of $-1, 1-\varepsilon, 1, 1+\varepsilon$. For small ε , the only sizable gap in the spectrum of A is the one of size $2-\varepsilon$ between the least two eigenvalues. The invariant subspaces corresponding to the eigenvalues on either side of the gap are

$$V_1 = \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}, \quad V_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}.$$

(In this hand-computed example, it's convenient to use eigenvectors for the columns of V_2 , but any other basis would also be fine.) Applying $V = [V_1 \ V_2]$ gives

$$A \leftarrow V^H A V = \left[\begin{array}{c|ccc} -1 & 0 & 0 & 0 \\ \hline 0 & 1-\varepsilon & 0 & 0 \\ 0 & 0 & 1+\varepsilon & 0 \\ 0 & 0 & 0 & 1 \end{array} \right], \quad B \leftarrow V^H B V = \left[\begin{array}{c|ccc} 1 & 0 & 0 & -\varepsilon \\ \hline 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -\varepsilon & 0 & 0 & 1 \end{array} \right].$$

Note that the entries in the bottom-left and top-right blocks of $V^H B V$ are on the order of ε . They are dropped:

$$B \leftarrow \left[\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

The problem has been broken into a trivial 1-by-1 subproblem in the top-left and a 3-by-3 subproblem in the bottom-right.

The bottom-right block of $V^H A V$ has three eigenvalues clustered near 1, but the bottom-right block of $V^H B V$ has eigenvalues $-1, 1, 1$. Because the second matrix has a larger spread, it is used to compute an invariant subspace, and the spectrum is split at the larger gap. We omit the details, but the next step ends with

$$A \leftarrow \left[\begin{array}{c|ccc} -1 & 0 & 0 & 0 \\ \hline 0 & 1 & \varepsilon & 0 \\ 0 & \varepsilon & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right], \quad B \leftarrow \left[\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

The off-diagonal entries of ε in A are discarded, and the final recursive step does not modify the matrices because they are already diagonal.

When the unitary transformations are accumulated, they produce

$$U = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \end{bmatrix},$$

and we conclude that

$$U^H A U \approx \text{diag}(-1, 1, 1, 1), \quad U^H B U \approx \text{diag}(1, -1, 1, 1).$$

Note that the algorithm avoided splitting any tight cluster of eigenvalues for the purpose of computing an invariant subspace. This is key to avoiding ill conditioning.

For a second example, consider the matrices

$$A = \begin{bmatrix} 1 & \varepsilon \\ \varepsilon & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \varepsilon & 0 \\ 0 & 1 + \varepsilon \end{bmatrix}.$$

Both matrices have eigenvalues $1 \pm \varepsilon$. What does the algorithm do with this small gap? It may split the spectrum of A at its center to obtain invariant subspaces

$$V_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}, \quad V_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix},$$

which bear no resemblance to the invariant subspaces of B . They are applied anyway:

$$V^H A V = \begin{bmatrix} 1 - \varepsilon & 0 \\ 0 & 1 + \varepsilon \end{bmatrix}, \quad V^H B V = \begin{bmatrix} 1 & \varepsilon \\ \varepsilon & 1 \end{bmatrix}.$$

Although these transformations provide no improvement, they are harmless. Even though V_1 and V_2 are poor approximations to B 's invariant subspaces, $V^H B V$ is nearly diagonal. The reason is this: because the operator represented by B has such a small spread, it nearly equals the identity regardless of the basis in which it is expressed.

The design and analysis of the spectral divide-and-conquer algorithm are motivated by examples like the two above: when there is a large eigenvalue gap, it is attacked; when only tiny gaps remain, the spread must be small, and the matrix is already nearly diagonal.

2.2 A pair of estimates

Our goal for the rest of this section is to prove backward stability of Algorithm 1, the spectral divide-and-conquer algorithm. We start by developing two bounds that are useful for reduction to block-diagonal form. They involve an eigenvalue gap and the spread of a matrix.

2.2.1 A residual bound. Suppose

$$A = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$$

is Hermitian, A_{11} is k -by- k and A_{22} is $(n - k)$ -by- $(n - k)$. Suppose

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

is unitary with blocks of the same size as A . Our concern is $Z^H A Z$, particularly its bottom-left block:

$$Z^H A Z = \begin{bmatrix} * & * \\ Z_{12}^H A_{11} Z_{11} + Z_{22}^H A_{22} Z_{21} & * \end{bmatrix}.$$

Under what circumstances is this block numerically small? One possibility is that Z is not far from the identity matrix. Then $Z^H A Z$ is not changed much from A , and its lower-left block stays close to zero. Another possibility is that A is close to a multiple of the identity matrix in the sense of having a small spread. In this case, $Z^H A Z$ also resembles a multiple of the identity, regardless of the choice for Z . Between these two extremes is a continuum that we wish to capture.

LEMMA 2.2 If A is block-diagonal and Z is unitary as above, then the lower-left block of $Z^H A Z$ satisfies

$$\left\| Z_{12}^H A_{11} Z_{11} + Z_{22}^H A_{22} Z_{21} \right\| \leq s(A) \|Z_{21}\|. \quad (2.2)$$

Proof. Let τ be the midpoint between the least and greatest eigenvalues of A , and let $X = \frac{2}{s(A)}(A - \tau I)$. Then

$$\|X\| = 1 \quad \text{and} \quad A = \tau I + \frac{s(A)}{2}X = \tau I + \frac{s(A)}{2} \begin{bmatrix} X_{11} & 0 \\ 0 & X_{22} \end{bmatrix}.$$

Thus

$$Z^H A Z = \tau I + \frac{s(A)}{2} Z^H X Z,$$

and the lower-left block equals

$$\frac{s(A)}{2} (Z_{12}^H X_{11} Z_{11} + Z_{22}^H X_{22} Z_{21}).$$

Noting that $\|Z_{12}\| = \|Z_{21}\|$ (a consequence of the CS decomposition; Paige & Wei (1994)) and that $\|Z_{11}\|, \|Z_{22}\|, \|X_{11}\|$ and $\|X_{22}\|$ are all ≤ 1 , we find that the norm of the lower-left block is bounded above as follows:

$$\frac{s(A)}{2} \|Z_{12}^H X_{11} Z_{11} + Z_{22}^H X_{22} Z_{21}\| \leq \frac{s(A)}{2} (\|Z_{21}\| \cdot 1 \cdot 1 + 1 \cdot 1 \cdot \|Z_{21}\|) = s(A) \|Z_{21}\|.$$

□

We earlier alluded to a bound based on two factors, one reflecting an eigenvalue gap and the other reflecting a spread. We see this taking shape in the right-hand side of (2.2). The factor $s(A)$ is the spread, and the factor $\|Z_{21}\|$ will soon be bounded in terms of an eigenvalue gap.

2.2.2 A corollary of the $\sin \theta$ theorem. The following lemma applies the $\sin \theta$ theorem (Appendix B) in a particular context. Let A and E be Hermitian matrices. Suppose the eigenvalues of A are $\lambda_1 \leq \dots \leq \lambda_n$ and that an eigenvalue gap $\lambda_{k+1} - \lambda_k > 0$ has been located. Let V_1 and V_2 identify invariant subspaces for the eigenvalues to the left and to the right of the gap, respectively. Let W_1 identify an invariant subspace for the k least eigenvalues of $A + E$, and let W_2 identify the orthogonal complement of W_1 . Adopt the notation $Z_{21} = V_2^H W_1$ from Theorem B.1.

LEMMA 2.3 Let $\gamma > 0$. If $s(A) \geq (2/\gamma)\|E\|$ and $\lambda_{k+1} - \lambda_k \geq \gamma s(A) > 0$, then

$$\|Z_{21}\| \leq \frac{2\|E\|}{\gamma s(A)}. \quad (2.3)$$

Proof. When A is perturbed to $A + E$, no eigenvalue moves by more than $\|E\|$ by Weyl's inequality (Horn & Johnson, 1990, p. 181). Therefore, the gap between the k least eigenvalues of A and the $n - k$ greatest eigenvalues of $A + E$ is at least $\gamma s(A) - \|E\| \geq \gamma s(A) - (\gamma/2)s(A) = (\gamma/2)s(A)$. Take this last value as δ in Theorem B.1. □

Note that the hypothesis $s(A) \geq (2/\gamma)\|E\|$ in the statement of the lemma is somewhat arbitrary; the coefficient $2/\gamma$ could be replaced by any quantity greater than $1/\gamma$ and the $\sin \theta$ theorem would still apply.

2.3 Reduction to block-diagonal form

We begin a step-by-step analysis of the spectral divide-and-conquer algorithm.

Let A and B be δ -commuting Hermitian matrices. Then

$$U^H(A + \Delta A)U = \Lambda, \quad U^H(B + \Delta B)U = M \quad (2.4)$$

for some unitary U and diagonal Λ, M and Hermitian $\Delta A, \Delta B$ with $\|\Delta A\|, \|\Delta B\| \leq \delta$. The purpose of the algorithm is to compute $\hat{U}$, $\hat{\Lambda}$ and $\hat{M}$ such that

$$\hat{U}^H A \hat{U} = \hat{\Lambda} \approx \Lambda, \quad \hat{U}^H B \hat{U} = \hat{M} \approx M$$

without knowledge of ΔA or ΔB .

Our focus in this section is on simultaneously transforming A and B to block-diagonal form—the responsibility of steps 1–4 in Algorithm 1.

Assume that

$$s(A) > 0, \quad s(A) \geq \beta s(B), \quad (2.5)$$

i.e., case 2 of step 1. In step 2, the algorithm locates an eigenvalue gap in the spectrum of A . Let $\hat{\lambda}_1, \dots, \hat{\lambda}_k$ be the eigenvalues of A to the left of the gap, and let $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_n$ be the eigenvalues to the right. The algorithm then computes invariant subspaces $\hat{V}_1$ and $\hat{V}_2$ for A that correspond to the two subsets of eigenvalues. The pair $\hat{V} = [\hat{V}_1 \ \hat{V}_2]$ of invariant subspaces block-diagonalizes A :

$$\hat{V}^H A \hat{V} = \begin{bmatrix} \hat{V}_1^H A \hat{V}_1 & 0 \\ 0 & \hat{V}_2^H A \hat{V}_2 \end{bmatrix}.$$

We wish to show that it nearly block-diagonalizes B as well.

Let $V = [V_1 \ V_2]$ contain invariant subspaces for the k least and $n - k$ greatest eigenvalues of $A + \Delta A$. (Our theoretical argument introduces V for the ‘ideal’ $A + \Delta A$, while the algorithm computes $\hat{V}$ for the ‘imperfect’ A .) We have

$$V^H(A + \Delta A)V = \begin{bmatrix} V_1^H(A + \Delta A)V_1 & 0 \\ 0 & V_2^H(A + \Delta A)V_2 \end{bmatrix}.$$

Because $A + \Delta A$ and $B + \Delta B$ commute, they have identical invariant subspaces, so V also reduces $B + \Delta B$:

$$V^H(B + \Delta B)V = \begin{bmatrix} V_1^H(B + \Delta B)V_1 & 0 \\ 0 & V_2^H(B + \Delta B)V_2 \end{bmatrix}. \quad (2.6)$$

Our goal is to show that $\hat{V}^H B \hat{V}$ is nearly block-diagonal. Considering (2.6), this would be true in particular if $\hat{V}$ were close to V (on the Grassmann manifold) and ΔB were small. To quantify the

difference between $\hat{V}$ and V , we introduce

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} := V^H \hat{V} = \begin{bmatrix} V_1^H \\ V_2^H \end{bmatrix} \begin{bmatrix} \hat{V}_1 & \hat{V}_2 \end{bmatrix}.$$

Note that

$$\hat{V}^H B \hat{V} = X - \hat{V}^H (\Delta B) \hat{V} \quad \text{with} \quad X = \hat{V}^H (B + \Delta B) \hat{V}, \quad (2.7)$$

and (2.6) implies

$$X = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} = Z^H \begin{bmatrix} V_1^H (B + \Delta B) V_1 & 0 \\ 0 & V_2^H (B + \Delta B) V_2 \end{bmatrix} Z. \quad (2.8)$$

Below, we bound the off-diagonal blocks of Z , X and $\hat{V}^H B \hat{V}$, in turn.

LEMMA 2.4 Let $0 < \alpha \leq 1$. Suppose $s(A) \geq (2n/\alpha)\delta$ and $\hat{\lambda}_1, \dots, \hat{\lambda}_k$ are separated from $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_n$ by a gap of width at least $(\alpha/n)s(A)$. Then

$$\|Z_{21}\| \leq \frac{2n\delta}{\alpha s(A)}. \quad (2.9)$$

Proof. Apply Lemma 2.3 with $E = \Delta A$ and $\gamma = \alpha/n$ and use the bound $\|\Delta A\| \leq \delta$ from δ -commutativity. $\square$

LEMMA 2.5 Let $0 < \alpha, \beta \leq 1$. Suppose $\hat{\lambda}_1, \dots, \hat{\lambda}_k$ are separated from $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_n$ by a gap of width at least $(\alpha/n)s(A)$. Then

$$\|X_{21}\| \leq \left(\frac{2n}{\alpha\beta} + 2 \right) \delta.$$

Proof. By (2.8),

$$X_{21} = Z_{12}^H \left(V_1^H (B + \Delta B) V_1 \right) Z_{11} + Z_{22}^H \left(V_2^H (B + \Delta B) V_2 \right) Z_{21}.$$

By Lemma 2.2,

$$\|X_{21}\| \leq s(V^H (B + \Delta B) V) \|Z_{21}\| = s(B + \Delta B) \|Z_{21}\|.$$

The first factor is bounded as follows using Weyl's inequality and (2.5):

$$s(B + \Delta B) \leq s(B) + 2\|\Delta B\| \leq \frac{s(A)}{\beta} + 2\delta.$$

The rest of the proof is in two cases. First, if $s(A) \geq (2n/\alpha)\delta$, then Lemma 2.4 gives

$$\|X_{21}\| \leq \left(\frac{s(A)}{\beta} + 2\delta \right) \frac{2n\delta}{\alpha s(A)} = \frac{2n}{\alpha\beta} \delta + \frac{4n\delta^2}{\alpha s(A)} \leq \frac{2n}{\alpha\beta} \delta + 2\delta.$$

Otherwise, $s(A) < (2n/\alpha)\delta$, and $\|Z_{21}\| \leq 1$ because Z is unitary, so

$$\|X_{21}\| \leq \left(\frac{s(A)}{\beta} + 2\delta \right) \cdot 1 < \frac{2n}{\alpha\beta} \delta + 2\delta.$$

□

LEMMA 2.6 Let $0 < \alpha, \beta \leq 1$. Suppose $\hat{\lambda}_1, \dots, \hat{\lambda}_k$ are separated from $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_n$ by a gap of width at least $(\alpha/n)s(A)$. Then the bottom-left block of $\hat{V}^H B \hat{V}$ is bounded as follows:

$$\|\hat{V}_2^H B \hat{V}_1\| \leq \left(\frac{2n}{\alpha\beta} + 3 \right) \delta. \quad (2.10)$$

Proof. By (2.7), the triangle inequality, norm invariance under unitary transformations and the previous lemma,

$$\|\hat{V}_2^H B \hat{V}_1\| \leq \|X_{21}\| + \|\Delta B\| \leq \left(\frac{2n}{\alpha\beta} + 2 \right) \delta + \delta.$$

□

Let's put things together. Suppose the spectral divide-and-conquer algorithm is run on Hermitian matrices A and B , at least one of which has positive spread. The following theorem shows that the matrices are simultaneously reduced to block-diagonal form leaving only small residuals.

THEOREM 2.7 Let A and B be δ -commuting Hermitian matrices. Steps 1–4 of the spectral divide-and-conquer algorithm (Algorithm 1) compute

$$\hat{V}^H A \hat{V} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} + E \quad (2.11)$$

and

$$\hat{V}^H B \hat{V} = \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix} + F \quad (2.12)$$

with

$$\|E\|, \|F\| \leq c_{n,\alpha,\beta} \delta. \quad (2.13)$$

Proof. Step 1 of the algorithm splits the problem into three cases.

The trivial case is that both matrices are multiples of the identity matrix. The algorithm immediately terminates and $E = F = 0$.

In the second case, given by (2.5), invariant subspaces are computed from A . Then $\|E\| \leq c_n^{(1)}\delta$ (see section 1.2.4) and

$$\hat{V}^H B \hat{V} = \begin{bmatrix} \hat{V}_1^H B \hat{V}_1 & 0 \\ 0 & \hat{V}_2^H B \hat{V}_2 \end{bmatrix} + G \quad \text{with} \quad G = \begin{bmatrix} 0 & (\hat{V}_2^H B \hat{V}_1)^H \\ \hat{V}_2^H B \hat{V}_1 & 0 \end{bmatrix}.$$

By the Jordan–Wielandt Theorem (Stewart & Sun, 1990, p. 32), $\|G\| = \|\hat{V}_2^H B \hat{V}_1\|$. Then Lemma 2.6 implies the bound $\|G\| \leq c_{n,\alpha,\beta}^{(2)}\delta$. Finally, the numerical computation of $\hat{V}^H B \hat{V}$ introduces an additional error of at most $c_n^{(3)}\delta$. (See again section 1.2.4.) Thus,

$$\|F\| \leq \|G\| + c_n^{(3)}\delta \leq c_{n,\alpha,\beta}^{(4)}\delta.$$

The theorem is proved with $c_{n,\alpha,\beta}$ equaling the greater of $c_n^{(1)}$ or $c_{n,\alpha,\beta}^{(4)}$.

In the third case, invariant subspaces are computed from B instead of A . The roles of A and B can be reversed in the above argument to find $\|F\| \leq c_n^{(1)}\delta$ and $\|E\| \leq c_{n,\alpha,\beta}^{(4)}\delta$. $\square$

Because $\hat{V}$ is unitary, the residuals can be thrown backward to become backward errors of the same norms. Thus, the reduction is backward stable, and the backward error is linear in δ .

REMARK 2.8 The analysis indicates that the choice of solver for the invariant subspace may have relatively little impact on the overall accuracy. The invariant subspace operations introduce errors that are proportional to unit roundoff u , as in (1.1). We expect that δ may sometimes be orders of magnitude greater than u , in which case the error bound from Lemma 2.6, which reflects the amelioration necessary to recover commutativity, dominates the analysis.

2.4 Recursion

After reduction to block-diagonal form, the smaller blocks are simultaneously diagonalized by recursively applying the same procedure.

There is a nontrivial concern: are the subproblems well posed? In the original problem, we must rely on the user to provide matrices A and B that are nearly commuting. However, it becomes the algorithm's responsibility to ensure that the submatrices passed to the recursive calls are nearly commuting.

By throwing the residuals E, F of Theorem 2.7 backward as described in the previous section, we see that small perturbations of the original input matrices A, B are simultaneously unitarily similar to block-diagonal matrices

$$\begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}, \quad \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix}.$$

We need to show that A_{11}, B_{11} and A_{22}, B_{22} are nearly commuting matrix pairs.

Because the original matrices nearly commute, there exist small backward errors ΔA_{ij} and ΔB_{ij} such that the following block matrices commute:

$$\begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} + \begin{bmatrix} \Delta A_{11} & \Delta A_{12} \\ \Delta A_{21} & \Delta A_{22} \end{bmatrix}, \quad \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix} + \begin{bmatrix} \Delta B_{11} & \Delta B_{12} \\ \Delta B_{21} & \Delta B_{22} \end{bmatrix}. \quad (2.14)$$

It is not immediately obvious that any of the submatrices must nearly commute.

THEOREM 2.9 Let

$$\begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}, \quad \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix}$$

be conformable block-diagonal Hermitian matrices that are δ -commuting. Then A_{11}, B_{11} are $(1 + \sqrt{k})\delta$ -commuting and A_{22}, B_{22} are $(1 + \sqrt{n-k})\delta$ -commuting, in which k and $n-k$ are the orders of the top-left and bottom-right blocks, respectively.

Proof. There exist $\Delta A = [\Delta A_{ij}]$ and $\Delta B = [\Delta B_{ij}]$ such that the block matrices in (2.14) commute and $\|\Delta A\| < \delta$ and $\|\Delta B\| < \delta$. In particular, the top-left block of the commutator $(A + \Delta A)(B + \Delta B) - (B + \Delta B)(A + \Delta A)$ equals zero:

$$(A_{11} + \Delta A_{11})(B_{11} + \Delta B_{11}) + \Delta A_{12}\Delta B_{21} - (B_{11} + \Delta B_{11})(A_{11} + \Delta A_{11}) - \Delta B_{12}\Delta A_{21} = 0.$$

This implies

$$(A_{11} + \Delta A_{11})(B_{11} + \Delta B_{11}) - (B_{11} + \Delta B_{11})(A_{11} + \Delta A_{11}) = -\Delta A_{12}\Delta B_{21} + \Delta B_{12}\Delta A_{21}$$

and therefore

$$\|(A_{11} + \Delta A_{11})(B_{11} + \Delta B_{11}) - (B_{11} + \Delta B_{11})(A_{11} + \Delta A_{11})\| \leq 2\delta^2 = \frac{2\varepsilon^2}{k-1}$$

with $\varepsilon = \sqrt{k-1}\delta$. By Theorem A.1, there exist $\Delta A'_{11}$ and $\Delta B'_{11}$ such that $A_{11} + \Delta A_{11} + \Delta A'_{11}$ and $B_{11} + \Delta B_{11} + \Delta B'_{11}$ commute and $\|\Delta A'_{11}\|, \|\Delta B'_{11}\| \leq \sqrt{k-1}\delta$. By the triangle inequality,

$$\|\Delta A_{11} + \Delta A'_{11}\|, \|\Delta B_{11} + \Delta B'_{11}\| \leq \delta + \sqrt{k-1}\delta \leq (1 + \sqrt{k})\delta.$$

The same argument applies to the lower-right blocks. □

In the recursive step, the spectral divide-and-conquer algorithm calls itself to simultaneously diagonalize A_{11} and B_{11} and then A_{22} and B_{22} . By Theorem 2.9, these are well-posed problems.

2.5 Reconstruction

In the following, we prove that the entire divide-and-conquer process of Algorithm 1 is backward stable:

THEOREM 2.10 The spectral divide-and-conquer algorithm computes a simultaneous diagonalization in a backward-stable fashion:

$$\hat{U}^H(A + \Delta A)\hat{U} = \hat{\Lambda}, \quad \hat{U}^H(B + \Delta B)\hat{U} = \hat{M}, \quad \|\Delta A\|, \|\Delta B\| \leq c_{n,\alpha,\beta}\delta.$$

Proof. Suppose A and B are δ -commuting. The proof is by induction on n , the number of rows and columns.

The base case is $n = 1$. The given scalars commute, and the algorithm returns them intact; $c_{1,\alpha,\beta} = 0$.

For the induction step, assume that the algorithm has been proved to be backward stable for all matrix sizes $< n$.

Theorem 2.7 shows that the reduction to block-diagonal form produces

$$\hat{V}^H(A + \Delta A^{(1)})\hat{V} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$$

and

$$\hat{V}^H(B + \Delta B^{(1)})\hat{V} = \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix}$$

with $\Delta A^{(1)} = -\hat{V}E\hat{V}^H$ and $\Delta B^{(1)} = -\hat{V}F\hat{V}^H$ satisfying $\|\Delta A^{(1)}\|, \|\Delta B^{(1)}\| \leq c_{n,\alpha,\beta}^{(1)}\delta$.

The computed block-diagonal matrices are $c_{n,\alpha,\beta}^{(2)}\delta$ -commuting by the triangle inequality, and by Theorem 2.9, the pairs A_{11}, B_{11} and A_{22}, B_{22} are both $c_{n,\alpha,\beta}^{(3)}\delta$ -commuting. The subproblems are solved to produce

$$\begin{aligned} \hat{U}_1^H(A_{11} + \Delta A_{11})\hat{U}_1 &= \hat{A}_1, & \hat{U}_1^H(B_{11} + \Delta B_{11})\hat{U}_1 &= \hat{M}_1, \\ \hat{U}_2^H(A_{22} + \Delta A_{22})\hat{U}_2 &= \hat{A}_2, & \hat{U}_2^H(B_{22} + \Delta B_{22})\hat{U}_2 &= \hat{M}_2, \end{aligned}$$

and by the induction hypothesis, $\|\Delta A_{11}\|, \|\Delta B_{11}\| \leq c_{k,\alpha,\beta}\delta$ and $\|\Delta A_{22}\|, \|\Delta B_{22}\| \leq c_{n-k,\alpha,\beta}\delta$. Let $c_{n,\alpha,\beta}^{(4)}$ be the maximum of $c_{j,\alpha,\beta}, j = 1, \dots, n-1$, so that all four bounds can be replaced by $c_{n,\alpha,\beta}^{(4)}\delta$.

To combine the subproblems, let

$$\hat{U} = \hat{V} \begin{bmatrix} \hat{U}_1 \\ \hat{U}_2 \end{bmatrix}$$

and

$$\Delta A = \Delta A^{(1)} + \hat{V} \begin{bmatrix} \Delta A_{11} & \\ & \Delta A_{22} \end{bmatrix} \hat{V}^H, \quad \Delta B = \Delta B^{(1)} + \hat{V} \begin{bmatrix} \Delta B_{11} & \\ & \Delta B_{22} \end{bmatrix} \hat{V}^H.$$

Then

$$\hat{U}^H(A + \Delta A)\hat{U} = \begin{bmatrix} \hat{A}_1 & 0 \\ 0 & \hat{A}_2 \end{bmatrix}$$

and

$$\hat{U}^H(B + \Delta B)\hat{U} = \begin{bmatrix} \hat{M}_1 & 0 \\ 0 & \hat{M}_2 \end{bmatrix},$$

and the triangle inequality bounds the norms of the backward errors by $c_{n,\alpha,\beta}^{(1)}\delta + c_{n,\alpha,\beta}^{(4)}\delta =: c_{n,\alpha,\beta}\delta$. $\square$

The divide-and-conquer algorithm is proved to be backward stable with an error proportional to δ .

3. Do-one-then-do-the-other (DODO) method

Our second numerical method is the ‘DODO’ method. This method was motivated by two related concerns over the divide-and-conquer method from the previous section:

1. Is a fast implementation possible, given the need for eigenvalue spreads and gaps at multiple stages of the computation?
2. In practice, what should be done about the unspecified behavior discussed in Remark 2.1? Specifically, is there an effective strategy for targeting eigenvalue gaps when multiple options exist according to the theory of the previous section?

To address the efficiency concern, our first thought was to amortize the cost of spread and gap calculations, calculating an entire spectrum initially and reusing it over several stages. This approach eventually developed into a modified algorithm rather than pure bookkeeping, which also allowed us to address the second concern above: when faced with multiple large eigenvalue gaps, split *all* of them.

Here is the idea: diagonalize one of the input matrices first. Then the spread and eigenvalue gaps are trivial to compute. Furthermore, this uncovers not just two complementary invariant subspaces V_1 and V_2 , but rather an invariant subspace for every isolated eigenvalue or eigenvalue cluster. For the recursive step, the problem is broken not into just two subproblems of roughly equal size, but rather into several subproblems, one for each isolated eigenvalue cluster. See Fig. 1 again.

This method is the do-one-then-do-the-other method (DODO) and is specified as Algorithm 2.

ALGORITHM 2 DODO (Do-one-then-do-the-other)

Input: Hermitian A and B ; $\alpha \in (0, 1]$.

Output: Unitary U such that $U^H A U$ and $U^H B U$ are approximately diagonal.

1. Compute lower bounds on the eigenvalue spreads: $s_-(A) \leq s(A)$ and $s_-(B) \leq s(B)$. See (3.1).
 2. Case 1: $s_-(A) \geq s_-(B)$.
 - (a) Compute an eigenvalue decomposition $A = U \Lambda U^H$ with ascending eigenvalues.
 - (b) Partition the eigenvalues in Λ into clusters in which λ_i and λ_{i+1} are in the same cluster if $\lambda_{i+1} - \lambda_i < (\alpha/n)s(A)$.
 - (c) Replace $A \leftarrow \Lambda$ and $B \leftarrow U^H B U$.
 - (d) Using recursion, simultaneously diagonalize the principal submatrices of A and B corresponding to each eigenvalue cluster. Accumulate the eigenvectors into U .
 3. Case 2: $s_-(A) < s_-(B)$.
 - (a) Run the same steps with the roles of A and B reversed.
-

Note that if every computed eigenvalue is isolated, then no recursion is necessary; the eigenvector matrix from one input is simply applied to the second input. In this case, simultaneous diagonalization is nearly as fast as a single eigenvalue decomposition. (And even when eigenvalues are highly clustered, the performance penalty is modest; see section 4.1.)

A number of additional points are addressed below.

3.1 Lower bound on the spread

As in the divide-and-conquer method, invariant subspaces are computed from one matrix or the other, and the matrix with greater eigenvalue spread is generally more reliable. However, when the spreads are comparable, there is little difference from choosing one over the other, and a rough, but rigorous estimate of the spread is sufficient.

There are a number of known lower bounds on the spread (Johnson *et al.*, 1985; Marcus & Minc, 1992; Merikoski & Kumar, 2003; Sharma *et al.*, 2015; Massey *et al.*, 2021), but we are not aware of any estimates with error bounds in the literature, so we develop our own.

Let X be an n -by- n Hermitian matrix and define $X_0 = X - \frac{\text{tr } X}{n}I$. Because the eigenvalues of X_0 lie within $\pm\|X_0\|$, we have $s(X_0) \leq 2\|X_0\|$. Because X_0 has its spectrum mean at zero, it must have at least one nonpositive and at least one non-negative eigenvalue, so $s(X_0) \geq \|X_0\|$. Together with the translation invariance of the spread, these bounds provide an order-of-magnitude estimate:

$$\|X_0\| \leq s(X) \leq 2\|X_0\|.$$

We can also incorporate the easier-to-compute Frobenius norm:

$$\frac{1}{\sqrt{n}}\|X_0\|_F \leq \|X_0\| \leq s(X) \leq 2\|X_0\| \leq 2\|X_0\|_F.$$

Therefore,

$$s_-(X) := \frac{1}{\sqrt{n}}\|X_0\|_F = \frac{1}{\sqrt{n}}\left\|X - \frac{\text{tr } X}{n}I\right\|_F \quad (3.1)$$

is an inexpensive lower bound on $s(X)$ that gives an order-of-magnitude estimate.

Suppose $s_-(A) \geq s_-(B)$ as in case 1 of Algorithm 2. Then

$$s(A) \geq s_-(A) \geq s_-(B) = \frac{1}{\sqrt{n}}\|B_0\|_F \geq \frac{1}{2\sqrt{n}}s(B).$$

Thus, the matrices satisfy the hypothesis $s(A) \geq \beta s(B)$ with $\beta = 1/(2\sqrt{n})$. Similarly, if $s_-(B) > s_-(A)$, then $s(B) \geq \beta s(A)$ with the same value of β .

3.2 Choice of eigenvalue solver

The decomposition $A = U\Lambda U^H$ can be computed using any eigenvalue routine. This makes the DODO algorithm adaptable to a wide range of computer architectures and allows it to improve automatically with advances for the symmetric eigenvalue problem.

3.3 Termination

The algorithm is guaranteed to terminate because there is always a sufficiently large gap to split the spectrum into two or more clusters, just as in the divide-and-conquer algorithm.

3.4 Backward stability

As an extension of the divide-and-conquer algorithm, the DODO algorithm is backward stable, and there are no surprises in the proof. The major changes in the algorithm are the following:

1. At each division stage, the matrices are reduced to $p \geq 2$ blocks instead of exactly 2 blocks.
2. Additional deterministic choices are made:
 - (a) β is set to $1/(2\sqrt{n})$, as discussed above;
 - (b) the potentially nondeterministic branch to step 1(b) or 1(c) of Algorithm 1 is made definite by the use of the spread estimate; and
 - (c) the spectrum is split at all eigenvalue gaps of sufficient size rather than an arbitrarily chosen one.

To prove backward stability, Theorems 2.7, 2.9 and 2.10 must be adapted. For reduction to block-diagonal form, a simple but sufficient analysis comes from applying Theorem 2.7 sequentially to each of the $p - 1$ eigenvalue gaps. The triangle inequality shows that the overall error is still proportional to δ . Theorem 2.9 can be adapted to show that A_{ii} and B_{ii} are $(1 + \sqrt{(p - 1)(k_i - 1)})\delta$ -commuting, in which k_i is the size of the i th diagonal block. Theorem 2.10 is essentially unchanged: the number of blocks along the diagonal is now p rather than 2, but otherwise the argument is the same.

All together, we have the following:

THEOREM 3.1 The DODO algorithm computes a simultaneous diagonalization in a backward-stable fashion, and the error term is proportional to δ .

3.5 Speed vs. stability

The parameter α helps to define the difference between a small eigenvalue gap and a large one. A smaller value of α permits the algorithm to break eigenvalue clusters more aggressively and terminate sooner, at the cost of increasing the error bound (2.10) reciprocally.

3.6 Summary

The DODO algorithm leverages an off-the-shelf eigenvalue solver to reduce both input matrices. Additional passes are necessary only when eigenvalue clusters are present, and these passes are relatively inexpensive because they involve smaller matrices. The entire process is proved to be backward stable.

4. Numerical experiments and applications

We ran the DODO (do-one-then-do-the-other) method through four different batteries of numerical tests.

4.1 Randomly generated test matrices

The first test suite involves a variety of randomly-generated matrix pairs exhibiting a range of eigenvalue clustering and deviation from commutativity, including some extremely difficult problems.

The test design begins with clustered eigenvalues. Let $x_1, \dots, x_n$ and $y_1, \dots, y_n$ be independent random variables drawn from the uniform distribution on $[-L, 0]$. Let $\lambda_i = \sum_{j=1}^i 10^{x_j}$ and $\mu_i =$

$\sum_{j=1}^i 10^{y_j}$. The following is a pair of spectra sampled from this distribution with $L = 20$:

0.000041765364255	0.000000000025468
0.114621464016922	0.000000000025506
0.114621464016948	0.701513892095242
0.114621464016974	0.701513913558071
0.114621464016974	0.747556026901336
0.114621485394550	0.747556031371058
0.114621485435445	0.747556031371076
0.114621485435445	0.747556031372738
0.318118381559489	0.747556031372738
0.403382694940690	0.747556031372738

With $L = 20$ as above, it is common to see repeated eigenvalues (when represented in double precision) and tight clusters of eigenvalues.

For the eigenvectors of the first matrix, we draw U uniformly at random from the space of n -by- n orthogonal matrices. For the second matrix, we let X be the antisymmetric part of a matrix with independent and identically-distributed standard Gaussian entries and note that $e^{(\gamma/2)X}$ is an orthogonal matrix that is close to the identity matrix when γ is small. Hence, $Ue^{(\gamma/2)X}$ is a perturbation of U that is still orthogonal.

The test matrices are

$$A = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^T, \quad B = (Ue^{(\gamma/2)X}) \operatorname{diag}(\mu_1, \dots, \mu_n) (Ue^{(\gamma/2)X})^T.$$

These are nearly commuting matrices with clustered eigenvalues. They are δ -commuting with

$$\delta = \|U \operatorname{diag}(\mu_1, \dots, \mu_n) U^T - B\| = O(\gamma).$$

Table 1 contains experimental results on stability. Each row of the table corresponds to a selection of values for n , L and γ . For each row, 100 trials were run with $\alpha = 0.01$, and the MATLAB `eig` routine was used to compute eigenvalue decompositions. To assess accuracy, the residuals

$$E_1 := \frac{\|\hat{A} - \hat{U}^T A \hat{U}\|}{\|A\| + \|B\|}, \quad E_2 := \frac{\|\hat{M} - \hat{U}^T B \hat{U}\|}{\|A\| + \|B\|}$$

and

$$\varepsilon := \|I - \hat{U}^T \hat{U}\|$$

were measured. We want the residuals E_1 and E_2 to be on the order of γ , indicating that the error is not much greater than the distance from commutativity, and ε to be on the order of unit roundoff, indicating that $\hat{U}$ is nearly orthogonal. The results in the table are encouraging, with E_1 and E_2 staying close to γ and ε remaining close to 10^{-15} . Note also that while n grows from $n = 10$ to $n = 1000$, a factor of 100, the ratios E_1/γ and E_2/γ grow by a factor of only about 10. (See the last two columns of the table.) This is the slow growth rate required by Problem 1.

TABLE 1 Accuracy data from random test cases

n	$[-L, 0]$	γ	$\max E_1$	$\max E_2$	$\max \varepsilon$	$(\max E_1)/\gamma$	$(\max E_2)/\gamma$
10	$[-5, 0]$	$1e-12$	$6e-13$	$5e-13$	$3e-15$	0.6	0.5
10	$[-20, 0]$	$1e-12$	$7e-13$	$6e-13$	$3e-15$	0.7	0.6
100	$[-5, 0]$	$1e-12$	$2e-12$	$2e-12$	$5e-15$	1.7	1.8
100	$[-20, 0]$	$1e-12$	$2e-12$	$2e-12$	$5e-15$	2.2	2.0
1000	$[-5, 0]$	$1e-12$	$5e-12$	$5e-12$	$1e-14$	5.4	5.2
1000	$[-20, 0]$	$1e-12$	$5e-12$	$5e-12$	$8e-15$	5.5	5.4
10	$[-5, 0]$	$1e-03$	$6e-04$	$7e-04$	$3e-15$	0.6	0.7
10	$[-20, 0]$	$1e-03$	$5e-04$	$5e-04$	$3e-15$	0.5	0.5
100	$[-5, 0]$	$1e-03$	$2e-03$	$2e-03$	$6e-15$	1.7	1.8
100	$[-20, 0]$	$1e-03$	$2e-03$	$2e-03$	$2e-14$	2.0	2.1
1000	$[-5, 0]$	$1e-03$	$5e-03$	$5e-03$	$1e-14$	5.3	5.3
1000	$[-20, 0]$	$1e-03$	$6e-03$	$5e-03$	$8e-15$	5.6	5.3

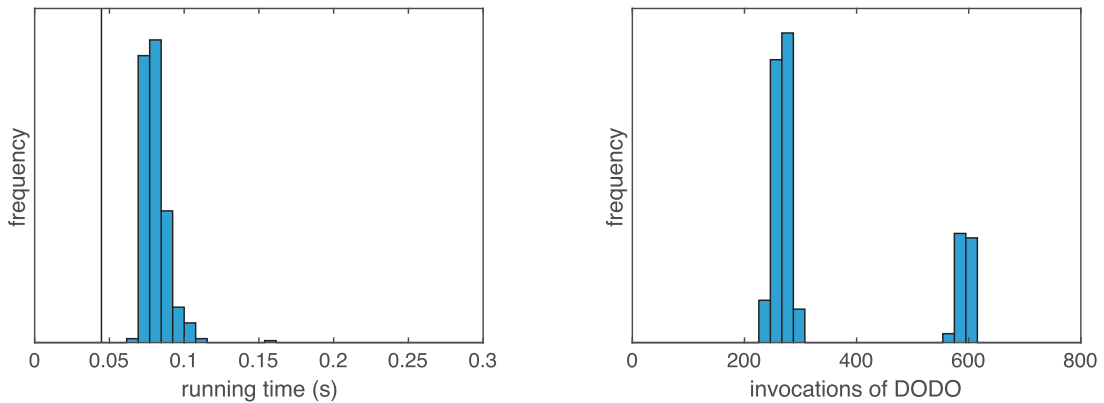


FIG. 2. Computational cost for the four test cases in Table 1 that involve 1000-by-1000 matrices.

Computational cost is considered in Fig. 2. The data come from the 100 trial runs of the four test cases of Table 1 involving matrices of size $n = 1000$. This is a heterogeneous data set, including matrices with tight and not-as-tight eigenvalue clusters. The histogram on the left shows running times measured in seconds from our MATLAB implementation running on a laptop computer, and the histogram on the right shows the number of recursive invocations of DODO. In the plot on the left, the vertical line indicates the median time consumed by the initial n -by- n eigenvalue computation. We make two observations: first, the running time is a small multiple of the cost of a single black-box eigenvalue-eigenvector computation, even on test cases with tightly clustered eigenvalues. Secondly, the running time varies little, even though the number of recursive invocations varies greatly. A closer inspection shows that the cases producing about 600 invocations of DODO are precisely the ones with $L = 20$ and $\gamma = 10^{-12}$ —those with the highest degree of eigenvalue clustering. Although these more difficult cases increase the number of invocations by a factor of about two or three over the other test cases, the increase in wall-clock running time is more modest because the additional executions involve relatively small matrices.

TABLE 2 Numerical data involving the diagonalization of several complex Hadamard matrices. MATLAB's *schur* is compared against the DODO method. Each matrix is measured by its Frobenius norm, and each measurement would ideally be zero

test case		schur			DODO		
n	k	$\text{im}Q$	$I - Q^H Q$	$\text{offdiag}(Q^H X Q)$	$\text{im}Q$	$I - Q^T Q$	$\text{offdiag}(Q^T X Q)$
3	1	1.3	8e-16	1e-15	0	8e-16	1e-15
4	1	1.2	1e-15	8e-16	0	8e-16	7e-16
5	1	1.5	1e-15	3e-15	0	1e-15	2e-15
6	1	1.6	1e-15	2e-15	0	1e-15	2e-15
6	3	1.4	1e-15	2e-15	0	1e-15	3e-15
7	1	1.1	2e-15	6e-15	0	1e-15	9e-15
8	1	1.7	2e-15	6e-15	0	2e-15	4e-15
9	1	2.1	4e-15	1e-14	0	2e-15	8e-15
10	1	2.3	5e-15	1e-14	0	3e-15	9e-15
11	2	1.8	4e-15	1e-14	0	3e-15	4e-14
12	1	1.9	4e-15	2e-14	0	3e-15	1e-14
13	1	2.7	2e-15	7e-15	0	2e-15	5e-15
13	2	2.6	2e-15	7e-15	0	2e-15	7e-15

4.2 Complex Hadamard matrices

Our second experiment involves the following structured eigenvalue problem (Horn & Johnson, 1990, p. 207):

THEOREM 4.1 If X is complex symmetric and normal, then there exist a real orthogonal matrix Q and a diagonal matrix Λ for which $X = Q\Lambda Q^T$.

The remarkable aspect is that Q is real orthogonal, not just complex unitary. A general purpose eigenvalue solver will typically produce a nonreal Q when the input X has nonreal entries. To compute a real orthogonal Q , we work with the real and imaginary parts of X separately, but in tandem (Bunse-Gerstner *et al.*, 1992): let

$$A = \text{re } X, \quad B = \text{im } X.$$

Then

$$Q^T A Q = Q^T \left(\frac{X + \bar{X}}{2} \right) Q = \frac{\Lambda + \bar{\Lambda}}{2} = \text{re } \Lambda$$

and

$$Q^T B Q = Q^T \left(\frac{X - \bar{X}}{2i} \right) Q = \frac{\Lambda - \bar{\Lambda}}{2i} = \text{im } \Lambda.$$

Thus, A and B are simultaneously diagonalized by the desired eigenvector matrix Q . Because A and B are real symmetric, Q has real entries as well.

TABLE 3 Numerical data involving the diagonalization of complex Hadamard matrices plus noise. The matrices are as in Table 2, except that Gaussian noise on the order of 10^{-6} has been added

test case		schur			DODO		
n	k	$\text{im}Q$	$I - Q^H Q$	$\text{offdiag}(Q^H X Q)$	$\text{im}Q$	$I - Q^T Q$	$\text{offdiag}(Q^T X Q)$
3	1	1.2	5e-16	2e-06	0	2e-16	2e-06
4	1	1.1	1e-15	4e-06	0	2e-15	4e-06
5	1	1.5	2e-15	3e-06	0	2e-15	3e-06
6	1	1.8	2e-15	5e-06	0	1e-15	5e-06
6	3	1.7	2e-15	5e-06	0	2e-15	7e-06
7	1	2.1	3e-15	7e-06	0	2e-15	1e-05
8	1	2.3	2e-15	5e-06	0	2e-15	5e-06
9	1	2.4	4e-15	1e-05	0	2e-15	9e-06
10	1	2.2	4e-15	1e-05	0	3e-15	9e-06
11	2	2.5	5e-15	1e-05	0	3e-15	3e-05
12	1	2.7	5e-15	1e-05	0	3e-15	1e-05
13	1	2.5	4e-15	1e-05	0	3e-15	9e-06
13	2	2.7	4e-15	1e-05	0	3e-15	1e-05

Some complex Hadamard matrices are symmetric and normal and have repeated eigenvalues as well. Hence, their real and imaginary parts make good test cases for the simultaneous diagonalization problem.

The Anymatrix library (Higham & Mikaitis, 2022) provides a collection of complex Hadamard matrices, each identified by a pair (n, k) of integers. We ran the DODO algorithm on the real and imaginary parts of every provided symmetric matrix of size 3-by-3 through 13-by-13 (eliminating examples of nonsymmetric matrices) and compared with MATLAB's `schur` routine. The results are in Table 2. Both routines effectively diagonalize the Hadamard matrix, but only DODO computes a real orthogonal solution (in the table, $\text{im}Q = 0$).

The test matrices in Table 2 have many repeated eigenvalues. Adding noise produces matrices with clusters of very close eigenvalues. In Table 3, we see results for matrices of the form $H + \varepsilon G$, in which H is a complex Hadamard matrix from the previous table and G is a symmetric matrix of independent random entries with real and imaginary parts from the standard Gaussian distribution. The Schur decomposition and DODO both leave noise in the off-diagonal part because the perturbed matrices are not quite normal. However, the DODO method is able to achieve its result with a real orthogonal similarity while the Schur decomposition uses a complex unitary similarity.

4.3 Symmetric Toeplitz matrix

Our third experiment involves a highly ill-conditioned eigenvector problem for a symmetric Toeplitz matrix.

Consider a Hermitian matrix with repeated or clustered eigenvalues. Perhaps, its ill-conditioned eigenvectors could be resolved with the help of another matrix with the same eigenvectors—in other words, a commuting Hermitian matrix. This idea has been explored specifically by pairing a symmetric Toeplitz matrix with a commuting tridiagonal matrix (Slepian, 1978; Grünbaum, 1981; Varah, 1993).

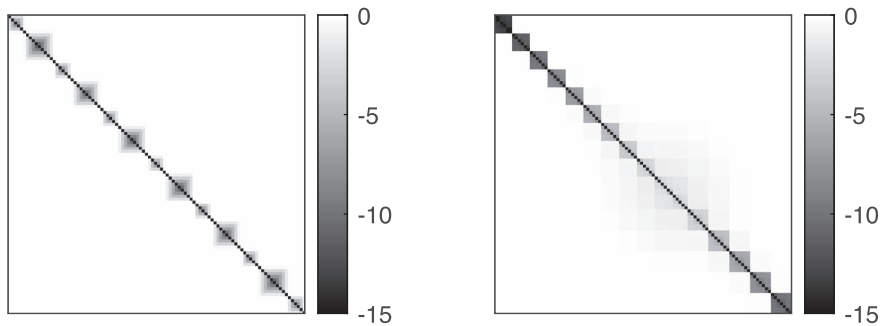


FIG. 3. Clustering in the spectra of commuting Hermitian matrices. Left: the symmetric Toeplitz matrix A ; right: the symmetric tridiagonal matrix B .

For a numerical experiment, we take from the literature the n -by- n symmetric Toeplitz matrix

$$A = [a_{ij}], \quad a_{ij} = \frac{\sin \alpha(i-j)}{\sin \beta(i-j)};$$

set $n = 100$, $\alpha = 0.1$ and $\beta = 0.2$; and construct a real symmetric tridiagonal matrix B that commutes with A (Grünbaum, 1981).

Both A and B have clustered eigenvalues and therefore ill-conditioned eigenvectors. Figure 3 illustrates this. Each plot is a heat map of $\log_{10}|\lambda_i - \lambda_j|$. Several eigenvalue clusters are evident in each spectrum from the darker shapes along the main diagonal; within these shapes, the eigenvalues are close together and the associated eigenvectors are numerically underdetermined. Note that because each spectrum is ordered separately, we cannot tell from these plots whether or how the eigenvalue clusters of A overlap with those of B .

As expected, a computed eigenvector matrix for A fails to diagonalize B and vice versa:

$$\begin{aligned} \|\text{offdiag}(U_A^T A U_A)\|_F &\approx 7 \cdot 10^{-13}, & \|\text{offdiag}(U_A^T B U_A)\|_F &\approx 1 \cdot 10^{-3}, \\ \|\text{offdiag}(U_B^T A U_B)\|_F &\approx 1 \cdot 10^1, & \|\text{offdiag}(U_B^T B U_B)\|_F &\approx 3 \cdot 10^{-14}. \end{aligned}$$

What we need is simultaneous diagonalization! DODO successfully computes

$$U^T A U \approx \Lambda, \quad U^T B U \approx M,$$

producing the residuals

$$\|U^T A U - \Lambda\|_F \approx 9 \cdot 10^{-13}, \quad \|U^T B U - M\|_F \approx 4 \cdot 10^{-12}.$$

Every column of U is thus constrained by two different condition numbers, one from A and one from B . If an eigenvector is well determined by *either* A or B , then it is well conditioned in the simultaneous diagonalization problem.

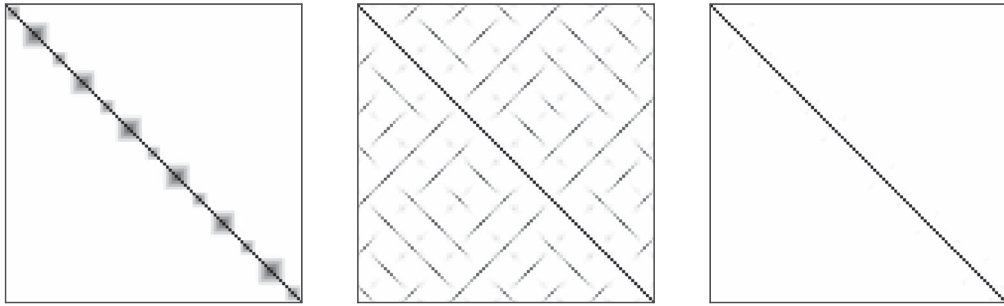


FIG. 4. The simultaneous diagonalization problem is well conditioned. Left and middle: eigenvalue clustering in A and B , with consistent eigenvector ordering; right: a combined clustering measure for the pair A, B .

Figure 4 indicates that the conditioning for this specific example is in fact quite good. Each plot explores the effect of swapping two columns of U . The plots on the left and in the middle show the residuals

$$\left\| (U\Pi_{ij})^T A (U\Pi_{ij}) - \Lambda \right\|_F$$

and

$$\left\| (U\Pi_{ij})^T B (U\Pi_{ij}) - M \right\|_F,$$

respectively, in which Π_{ij} is the permutation matrix that swaps columns i and j . Dark portions indicate pairs i, j of eigenvectors that can be swapped with little harm because they belong to eigenvalues that are nearly equal. These graphics are basically equivalent to the ones in the previous figure, but now the ordering is by a common eigenvector matrix. We see that the eigenvalue clusters are arranged quite differently between the two matrices. The plot on the right shows the combined residual

$$\left\| (U\Pi_{ij})^T A (U\Pi_{ij}) - \Lambda \right\|_F + \left\| (U\Pi_{ij})^T B (U\Pi_{ij}) - M \right\|_F.$$

It shows no common clusters and indicates that the simultaneous diagonalization problem is much better conditioned than either individual eigenvalue problem.

DODO has computed every eigenvector accurately.

4.4 A fixed point of the Jacobi-like method

The fourth test case is one on which the classic Jacobi-like method fails to converge without additional intervention. Bunse-Gerstner *et al.* credit the example to Voevodin (Voevodin & Ikramov, 1967; Bunse-Gerstner *et al.*, 1993). Let $n = 7$ and

$$A = \begin{bmatrix} a_{ij} \end{bmatrix}, \quad a_{ij} = \begin{cases} \cos((i+j)\pi/n) & i \neq j, \\ (1 - \frac{n}{2}) \cos(2i\pi/n) & i = j, \end{cases}$$

$$B = \begin{bmatrix} b_{ij} \end{bmatrix}, \quad b_{ij} = \begin{cases} \sin((i+j)\pi/n) & i \neq j, \\ (1 - \frac{n}{2}) \sin(2i\pi/n) & i = j. \end{cases}$$

The Jacobi iteration has a fixed point at this pair of commuting matrices and therefore does not converge. DODO successfully diagonalizes the pair, achieving

$$\|I - U^T U\|_F \approx 1 \cdot 10^{-15}, \quad \|U^T A U - \Lambda\|_F \approx 3 \cdot 10^{-15}, \quad \|U^T B U - M\|_F \approx 3 \cdot 10^{-15}.$$

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Data and code availability

A MATLAB implementation of the DODO method is available at <https://github.com/briansutton/dodo>. The repository also contains code for running the experiments of section 4 and generating data.

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A. Nearly commuting versus almost commuting

Nearly commuting and *almost commuting* are related, but different notions.

Matrices A and B *nearly commute* if nearby matrices exactly commute, i.e., if there exist ΔA and ΔB of small norm such that $A + \Delta A$ and $B + \Delta B$ commute. If A and B nearly commute and $A, B, \Delta A$ and ΔB are Hermitian, then A and B are nearly simultaneously diagonalizable. Specifically, there exist a unitary U and diagonal Λ and M such that

$$U^H(A + \Delta A)U = \Lambda, \quad U^H(B + \Delta B)U = M.$$

Our algorithm is designed for matrices that nearly commute in this sense.

The alternative notion is this: matrices A and B *almost commute* if their commutator $AB - BA$ is small in norm.

There is a substantial and still-growing literature, instigated in part by a 1969 note by Rosenthal in the *American Mathematical Monthly* (1969), on the relationship between the notions of near and almost commutativity. A quick proof shows that if A and B nearly commute with $\|\Delta A\|$ and $\|\Delta B\|$ on the order of ε , then their commutator is also $O(\varepsilon)$. What about the converse? For us, the most important result is the following theorem from 1979. (Actually, the original statement is a bit stronger.)

THEOREM A.1 (Percy & Shields, 1979). Let A and B be Hermitian matrices. If

$$\|AB - BA\| \leq \frac{2\varepsilon^2}{n-1},$$

then there exist Hermitian ΔA and ΔB such that $A + \Delta A$ and $B + \Delta B$ commute and $\|\Delta A\| \leq \varepsilon$ and $\|\Delta B\| \leq \varepsilon$.

The theorem suggests that a small residual $AB - BA$ may not imply that A and B lie so near a commuting matrix pair. This is indeed possible. Consider

$$A = \begin{bmatrix} 1 & \sqrt{\varepsilon} \\ \sqrt{\varepsilon} & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \sqrt{\varepsilon} & 0 \\ 0 & 1 + \sqrt{\varepsilon} \end{bmatrix}.$$

The commutator is

$$AB - BA = \begin{bmatrix} 0 & 2\varepsilon \\ -2\varepsilon & 0 \end{bmatrix},$$

of norm 2ε . The matrices are within $O(\sqrt{\varepsilon})$ of commuting matrices—just move both to the identity matrix—but not within $O(\varepsilon)$: an eigenvector matrix for $A + \varepsilon C$ is

$$\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} + O(\sqrt{\varepsilon})$$

and an eigenvector matrix for $B + \varepsilon D$ is

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + O(\sqrt{\varepsilon}),$$

for Hermitian matrices C and D of bounded norm. For sufficiently small ε , no perturbations of A and B on the order of ε are capable of bringing the eigenvectors into alignment. Thus, a commutator of norm ε does not imply the existence of commuting matrices at a distance of $O(\varepsilon)$.

Despite the relative weakness of the bound in Theorem A.1, it is still useful in the proof of Theorem 2.9. The lesson is that we can establish backward errors ΔA and ΔB of $O(\varepsilon)$ by showing a commutator of $O(\varepsilon^2)$.

B. The $\sin \theta$ theorem

The $\sin \theta$ theorem of Davis and Kahan (1969; 1970) concerns the perturbation theory of invariant subspaces of Hermitian matrices.

Let A be a Hermitian matrix. Let V_1 be a matrix whose columns form an orthonormal basis for an invariant subspace of A , and let V_2 identify its orthogonal complement. The $\sin \theta$ theorem bounds the perturbation of the invariant subspace V_1 induced by a perturbation of A .

Let E be another Hermitian matrix, in application often having numerically small entries. Let W_1 and W_2 be complementary invariant subspaces for $A + E$.

The following theorem is a corollary of the $\sin \theta$ theorem.

THEOREM B.1 Suppose the spectrum of $V_1^H A V_1$ lies in an interval separated from the spectrum of $W_2^H (A + E) W_2$ by distance $\delta > 0$. Then

$$\|V_2^H W_1\| = \|V_1^H W_2\| \leq \frac{\|E\|}{\delta}.$$

When $V_2^H W_1$ is small, W_1 is nearly orthogonal to V_2 and hence is just a small rotation away from the original V_1 . According to the theorem, this is guaranteed when the perturbation E is small and there is a reasonably large gap between the relevant portions of the spectra of A and $A + E$.

Another useful way to think about the $\sin \theta$ theorem is through the actions of the perturbed invariant subspaces. Because V_1 and V_2 identify invariant subspaces, we have

$$V^H A V = \begin{bmatrix} V_1^H A V_1 & 0 \\ 0 & V_2^H A V_2 \end{bmatrix},$$

in which $V = \begin{bmatrix} V_1 & V_2 \end{bmatrix}$. Similarly,

$$W^H(A+E)W = \begin{bmatrix} W_1^H(A+E)W_1 & 0 \\ 0 & W_2^H(A+E)W_2 \end{bmatrix}$$

with $W = \begin{bmatrix} W_1 & W_2 \end{bmatrix}$. Let $Z = V^H W$. Applying the perturbed eigenvectors of $A + E$ to the original matrix A gives

$$W^H A W = Z^H (V^H A V) Z.$$

The matrix Z reveals the extent to which the block-diagonal structure is preserved. If $Z_{21} = V_2^H W_1$ and $Z_{12} = V_1^H W_2$ are small, then Z is nearly block-diagonal and therefore $W^H A W$ is also nearly block-diagonal. The $\sin \theta$ theorem bounds the off-diagonal blocks of Z .